

The Augmented Synthetic Historical Control Method

Introduction

Much of economic and policy research in causal inference relies on comparing a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics for estimating treatment effects, second only to difference-in-differences. The central idea of SCM is straightforward: the untreated potential outcomes of a treated unit can be approximated by a weighted average of donor units. Difference-in-Differences designs operate on a similar principle, under the assumption of parallel pre-intervention trends [Baker et al., 2025].

However, suitable donor units may not always exist—for instance, during global events (e.g., pandemics, financial crises) or when no valid donor days are available. To address such cases, Chen et al. [2024] introduced the *Synthetic Historical Control* method (SHC). Instead of relying on external donors, SHC constructs a donor pool from the treated unit’s own pre-treatment history, splitting the pre-treatment period into overlapping segments. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, based on the assumption that future outcomes resemble past patterns. Like Abadie et al. [2010, 2015], SHC restricts weights to the simplex, ensuring convex combinations of donor segments. While this guarantees interpretability, it can be overly restrictive in time-series settings where noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the *Augmented Synthetic Historical Control* (ASHC) method, which extends SHC in two key ways.

First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, enabling controlled extrapolation beyond the convex hull of observed segments in exchange for better fit.

Second, ASHC introduces a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM of Ben-Michael et al. [2021]. This encourages the estimator to remain close to SHC unless extrapolation is justified. Importantly, ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where pre-treatment fit would otherwise be imperfect.

We establish the theoretical properties of ASHC. We derive a finite-sample error bound, decomposing the prediction error into four interpretable components: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. We then prove that ASHC is asymptotically consistent: as pre-treatment length and smoothing order grow, the error converges to the noise floor, and under sub-Gaussian noise, to zero in probability. Finally, we obtain the convergence rate, showing that under the optimal bandwidth, the error decays at $O_p(m^{-H/(2H+3)})$, approaching the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order increases.

The remainder of the paper is structured as follows. Section 2 introduces the assumptions and implementation of the SHC method, followed by the ASHC method. Section 3 presents the finite-sample and asymptotic analysis. Section 4 reports calibrated simulations, and Section 5 provides an empirical application. Proofs are given in the appendix. Implementation is carried out using Python’s `mlsynth` library (<https://mlsynth.readthedocs.io/en/latest/>).

References

- Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of california’s tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010. URL <https://doi.org/10.1198/jasa.2009.ap08746>.
- Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2):495–510, 2015. doi: <https://doi.org/10.1111/ajps.12116>.
- Andrew Baker, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H. C. Sant’Anna. Difference-in-differences designs: A practitioner’s guide, 2025. URL <https://arxiv.org/abs/2503.13323>.
- Eli Ben-Michael, Avi Feller, and Jesse Rothstein. The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536):1789–1803, dec 2021. doi: 10.1080/01621459.2021.1929245.
- Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: <https://ssrn.com/abstract=4995085>, September 2024. URL <http://dx.doi.org/10.2139/ssrn.4995085>.